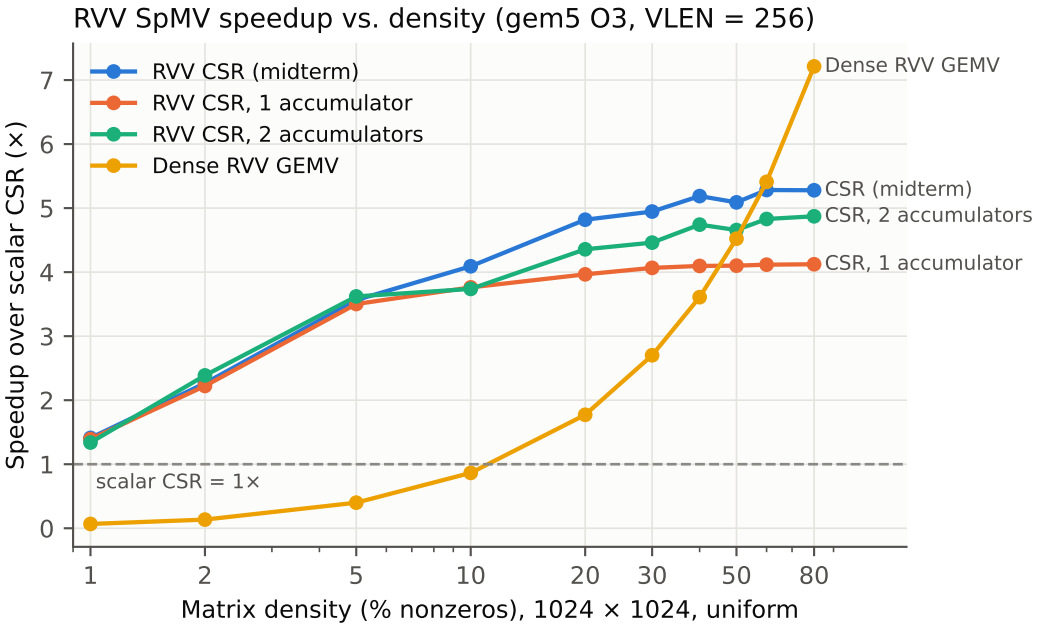
RVV SpMV speedup vs. density (gem5 O3, VLEN = 256)
RVV CSR (midterm)
RVV CSR, 1 accumulator
RVV CSR, 2 accumulators
Dense RVV GEMV
Dense RVV GEMV
CSR (midterm)
CSR, 2 accumulators
CSR, 1 accumulator
scalar CSR = 1×
Speedup over scalar CSR (×)
0
1
2
3
4
5
6
7
1
2
5
10
20
30
50
80
Matrix density (% nonzeros), 1024 × 1024, uniform